RESPONSE TO A REQUEST FOR PROPOSAL · ILLUSTRATIVE

Agentic and generative AI across the payments value chain

Assessment framework, prioritisation and a 24-month roadmap, with a worked example for Bank X, a large Indian private-sector bank

Payments & Fintech Practice · Practice Lead's response · October 2026

THE PAYMENTS VALUE CHAIN WE ASSESS, STAGE BY STAGE

1. Onboard
2. Initiate
3. Authenticate
4. Screen
5. Process
6. Settle
7. Reconcile
8. Resolve
9. Serve
10. Comply

How to read this document. It is written as a consulting practice lead would answer an RFP from an Indian bank: our understanding of the brief, a point of view, the assessment framework, a fully worked example with a prioritised portfolio and value case, deep dives, roadmap and the proposed engagement.

Illustrative document for learning. Bank X is fictional; every bank-specific number is an assumption, labelled A. Market and regulatory facts are cited (as of 13 Oct 2026). Not produced by or affiliated with any named firm.

EXECUTIVE SUMMARY ILLUSTRATIVE

Bank X can unlock about ₹465 crore a year by Year 3 by rewiring its highest-cost payment journeys with agentic AI, governed to RBI's emerging standard

1. **The opportunity sits in operations, not in chatbots.** Five domains (financial crime, disputes, customer and merchant servicing, cards, and transaction banking with cross-border) hold 70% of the value. They combine high handling cost, rule-bound decisions and unstructured evidence, which is where agentic AI does best.
2. **Five lenses decide every opportunity, in a fixed order.** Where it sits in the value chain; which kind of AI fits (fraud scoring stays classic ML); how much autonomy is safe; a value, feasibility and fit score behind six risk gates; then portfolio balance and sequencing.
3. **Regulation shapes the design, not just the sign-off.** Payment data stays in India. RBI's FREE-AI framework and its June 2026 draft model-risk guidance point to board-approved AI governance, model inventories, human oversight and kill switches. DPDP obligations apply from May 2027.
4. **Sequence for learning and trust: three waves over 24 months.** Wave 1 builds the platform and internal agents at supervised autonomy (disputes, AML triage, reconciliation, merchant onboarding and four copilots). Wave 2 goes customer- and merchant-facing. Wave 3 invents new revenue and adds collections once conduct controls are proven.
5. **The economics hold even if value disappoints.** About ₹180 crore one-time and ₹55 crore a year to run. Cumulative breakeven in month 17; month 20 if only 70% of the value arrives.
6. **Start with a 12-week diagnostic and two pilots.** Validate the value on Bank X's own data, build two sandbox pilots (UPI disputes, AML triage) and decide go or no-go at week 12. Indicative fee ₹6.9 crore against ₹465 crore a year at stake.

₹465 cr

annual value at stake by Year 3 (range ₹350–580 cr), C

50 → 21 → 10

use cases identified → scored → prioritised

Month 17

cumulative breakeven on ₹180 cr one-time investment, C

3 waves

over 24 months; supervised autonomy first

Labels: P = public source, A = assumed for illustration, C = calculated from P and A.

WHY THIS APPROACH

Domains, not demos
Value sized activity by activity on Bank X's own data (p. 12), not from vendor benchmarks.

Safe by design for RBI
Six risk gates and an autonomy cap are applied before anything is ranked (pp. 13–15).

Value proven in 12 weeks
Two sandbox pilots with evaluation thresholds and a go or no-go at week 12 (pp. 39–40).

Value is 'at stake' and to be validated in the diagnostic, not promised. Market facts cited on p. 48. · Bank X is a fictional composite; bank figures are assumptions.

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Reading paths.

Ten minutes: executive summary, p. 10 (framework), p. 23 (matrix), p. 32 (roadmap).

For the RFP panel: Sections 1, 3 and 7, plus the compliance matrix on p. 43.

For practitioners learning the method: Sections 3 and 4 and the appendix on p. 45.

Bank X has asked for a value-led, regulator-ready plan for generative and agentic AI across its payments business

Objectives, as indicated in the RFP

O1	Identify where generative and agentic AI can create value across the end-to-end payments value chain: UPI and retail payments, cards, merchant acquiring, transaction banking, cross-border and trade.
O2	Provide a transparent assessment framework and a prioritised, sequenced portfolio.
O3	Quantify value at stake, investment and payback for the priority use cases.
O4	Align with RBI, NPCI and data-protection requirements, including data localisation and model risk.
O5	Define the target AI architecture, operating model and governance.
O6	Recommend a 24-month roadmap, with two or three pilots started during the engagement.

Why now, as indicated: UPI volumes above 24 billion a month, a new merchant discount rate on larger UPI merchant payments from 15 October 2026, rising scam losses, and peers piloting agentic payments. The bank wants to act before cost-to-serve and fraud outgrow the operations model.

Scope

IN SCOPE

Payment products and operations; servicing and collections for cards; financial-crime operations for payments; payments technology and engineering.

OUT OF SCOPE

Lending decisions (except merchant credit built on payments data); treasury and markets; branch operations; vendor selection (we stay vendor-neutral and provide evaluation criteria).

Open questions for Bank X

1. What level of customer-facing autonomy is the board comfortable with in the first 12 months?
2. Which AI platforms, cloud regions and vendor contracts are already committed?
3. Can a secure sandbox with masked production data be ready by week 3?
4. Is Bank X willing to join NPCI's agentic-payments pilots as an issuer or acquirer?
5. Is there an investment envelope or a payback hurdle for the programme?

The RFP is assumed for this illustration; objectives are paraphrased as a bidder would play back a real brief. · Bank X is a fictional composite; bank figures are assumptions.

Behind the RFP sit five questions the board needs answered, and our approach is built around them

How should Bank X use generative and agentic AI to create value in payments, safely and at scale?

1. Where is the value?	Which payment journeys carry the most cost, loss and revenue upside, and how much of that work can AI do?	Lens 1 · value pools by domain
2. What is feasible now?	Do the data, systems, skills and AI techniques exist to deliver within 6–12 months?	Lens 2 and 4 · archetype fit, feasibility score
3. What is safe and compliant?	Which decisions can an agent take, under what limits, and does it meet RBI, NPCI and DPDP expectations?	Lens 3 and risk gates · autonomy caps
4. What first, what next?	Which use cases go in which wave, and what platform must exist first?	Lens 5 · portfolio, dependencies, waves
5. How do we scale and sustain?	What architecture, operating model, governance and value tracking will make it stick?	Sections 6 and 7

Each branch maps to a lens of the framework (Section 3) and to a deliverable of the engagement (Section 7).

Indian payments are growing fast while margins thin and fraud and regulatory scrutiny tighten

24.07 bn

UPI transactions in September 2026, worth ₹29.37 lakh crore: about 802 million a day.

NPCI, via Business Standard and Entrackr (P)

0.4%

Merchant discount rate on person-to-merchant UPI payments above ₹2,000, capped at ₹300, from 15 Oct 2026; small merchants exempt.

Press reports; verify against NPCI and RBI notices (P)

12.4 crore

Credit cards outstanding in August 2026, with ₹2.02 lakh crore of spend that month.

RBI data, via Business Standard (P)

8 · 29

Banks live on RBI's DPIP fraud Smart Registry (1 Sept 2026) · banks using RBIH's MuleHunter.AI (July 2026).

MediaNama; Finance Ministry reply via Taxguru (P)

7 · 6 · 26

FREE-AI sutras, pillars and recommendations (RBI, Aug 2025); draft model-risk guidance covering AI issued 24 June 2026.

KPMG India; S&R Associates (P)

Agents pay

NPCI–OpenAI–Razorpay pilot let ChatGPT pay over UPI within user limits (Oct 2025); NPCI's Unified Agent Protocol reported Sept 2026.

MediaNama; Startupfeed (P)

So what for Bank X. Volumes grow 20% or more a year while unit revenue stays thin, and fraud and regulatory scrutiny rise. The bank cannot staff its way through this. AI that *does the work* (agents that gather evidence, reconcile and resolve), not only AI that answers questions, is the lever that bends the cost curve, provided it is governed to RBI's emerging standard.

SIGNALS FROM THE MARKET (P)

NPCI · MyUPI
Conversational AI assistant inside BHIM for complaints and payment queries in five languages (Sept 2026).

BharatPe
Launched an agentic AI platform for its merchants, as reported in the trade press (2026).

NPCI · AiNxt
NPCI's suite of AI-led products for the UPI ecosystem, as reported; scope for member banks to be confirmed.

Axis Bank · Airtel Payments Bank
Bank partners in the first UPI agentic-payments pilot with OpenAI and Razorpay (Oct 2025).

Facts as reported in public sources on or before 13 Oct 2026 (full list p. 48). Some figures come from secondary reporting of official releases.

Not every payments problem needs generative AI: we match each one to one of six AI archetypes

	Archetype	What it does	Payments examples	Typical autonomy	Maturity
A	**Predictive ML**	Scores, ranks or forecasts from structured data	Fraud and mule scores, routing, liquidity forecasts, false-decline reduction	L3–L4	Mature
B	**GenAI copilot**	Drafts, summarises and searches knowledge for staff	Agent assist, policy Q&A, STR narrative drafts, code and test generation	L1	Mature
C	**Document AI**	Reads unstructured documents and messages	KYB documents, LC packs, remittance advice, ISO 20022 enrichment	L2	Maturing
D	**Conversational**	Talks to customers and merchants in Indian languages	Servicing assistant, merchant help, collections reminders	L2–L3	Maturing
E	**Agentic workflow**	Plans and executes multi-step work across systems using tools	Disputes, reconciliation breaks, payment repair, onboarding	L2 → L3	Emerging
F	**Agentic commerce**	External AI agents initiate payments for customers	Delegated mandates, agent tokens, agent verification	L3 within mandate	Nascent

Rule we apply: use the simplest archetype that does the job. Real-time fraud scoring stays classic ML (A), because it is faster, cheaper and easier to validate; generative AI adds value around it, in investigation (B) and customer interdiction (D).

Where Wave 1 value comes from: archetypes B, C and E inside operations, behind the bank's own staff. Customer-facing archetypes (D, F) follow once evaluation and controls are proven.

THREE MISTAKES WE SEE IN BANK AI PORTFOLIOS

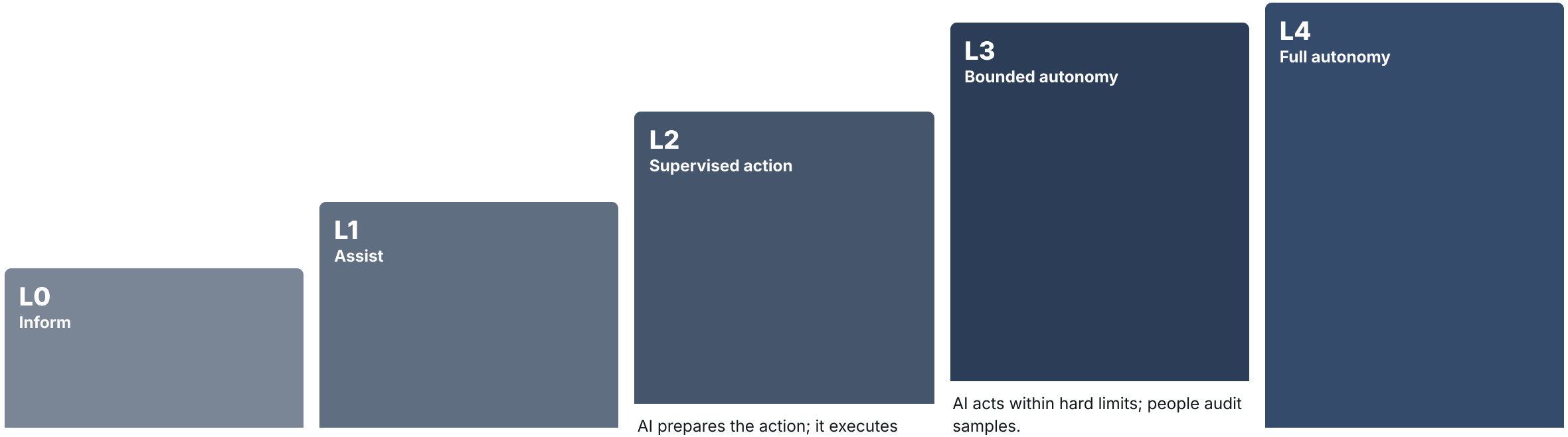
1 · An LLM where a rule would do
Fee reversals or TAT checks are deterministic. A language model adds cost, latency and validation burden with no gain.

2 · A chatbot without tools
An assistant that can only explain policy hands the customer back to a queue. Value needs actions: raise a dispute, block a card.

3 · An agent without evaluation
No golden test set, no graders, no thresholds: the bank cannot show the regulator, or itself, that the agent is safe to scale.

Archetypes and autonomy levels are the practice's working taxonomy; maturity reflects deployment evidence in banking as of 2026.

Autonomy is a design choice: we set a cap for each use case before we score it



AI produces insight; a person decides and acts.

EXAMPLE
Merchant insights dashboard

CONTROLS
Accuracy checks

AI drafts or recommends; a person acts.

EXAMPLE
STR narrative draft; agent assist

CONTROLS
Review of every output

AI prepares the action; it executes only after approval.

EXAMPLE
Dispute credit proposed, officer approves

CONTROLS
Maker-checker, full audit trail

AI acts within hard limits; people audit samples.

EXAMPLE
Auto-resolve UPI failed-debit disputes under ₹5,000

CONTROLS
₹ and volume limits, live monitoring, kill switch

AI acts end to end; people handle exceptions.

EXAMPLE
Payment routing and retries

CONTROLS
SLAs, drift monitoring, rollback

Red line for Bank X: decisions that block customer funds, refuse a dispute, deny credit or file a regulatory report stay at L2 or below until model-risk validation and board-approved AI governance are in place.

Autonomy levels adapted from common agent-autonomy taxonomies to payments controls (maker-checker, limits, audit).

We go in with five hypotheses, and the diagnostic is designed to prove or kill each one

	Hypothesis	Why we believe it	How the diagnostic tests it	If it fails
H1	**Value concentrates in operations-heavy domains (disputes, reconciliation, financial crime), not in chatbots.**	These domains combine high FTE cost, rule-bound work and unstructured evidence.	Activity-based time sampling in three domains; volumes from case systems.	Servicing moves into Wave 1
H2	**Value comes from rewiring end-to-end domains, not from isolated use cases.**	BCG reports 30–50% efficiency for 'reshape' plays versus about 10–15% for 'deploy' tools.	Compare domain-level and point-solution value cases for disputes.	Fund point copilots first; delay domain redesign
H3	**Data localisation and model risk will shape the architecture: India-hosted models and an LLM gateway are prerequisites.**	RBI's 2018 storage directive; the June 2026 draft model-risk guidance.	Architecture review with the CISO and CRO in weeks 2–4.	Customer-facing use cases can start earlier
H4	**Classic ML stays the backbone of fraud detection; generative AI adds investigation speed and customer interdiction.**	Scam losses come from authorised payments that rules and scores miss.	Decompose 12 months of fraud losses by typology.	Re-weight value towards fraud models
H5	**Most of the effort is people and process change, not models.**	BCG's 10-20-70 rule: 10% algorithms, 20% technology and data, 70% people and processes.	Change-readiness assessment of three operations teams.	Cut the change budget; reinvest in platform

A hypothesis that fails changes the plan, not just the slide. Each is tested by week 4 and reported at the first steering committee, before any scoring starts.

BCG, 'CEO's guide to maximizing value from AI' (July 2024); RBI, Storage of Payment System Data (April 2018); RBI draft model-risk guidance (June 2026).

We assess every opportunity through five lenses, always in the same order

LENS 1 **Where to play**	LENS 2 **What kind of AI**	LENS 3 **How much autonomy**	LENS 4 **Is it worth it**	LENS 5 **When and how**
Map 10 value-chain stages × 5 business lines; size cost, loss and revenue pools by activity.	Assign one of six archetypes; flag work that needs no AI or only classic ML.	Set the autonomy cap from error cost, reversibility and regulation; design human checkpoints.	Pass six risk gates, then score value, feasibility and strategic fit on anchored 1–5 scales.	Balance deploy, reshape and invent; respect dependencies; sequence into waves.
OUTPUT **Value-pool heatmap; domain list**	OUTPUT **Archetype per use case**	OUTPUT **Autonomy cap and control set**	OUTPUT **Priority index; risk tier**	OUTPUT **Roadmap; business case**

ONE USE CASE THROUGH THE FIVE LENSES: S01 DISPUTE RESOLUTION AGENT

Stage 8 · Resolve; UPI, cards and acquiring. ₹68 cr a year at stake.	Archetype E · agentic workflow: multi-step evidence gathering across 6–8 systems.	Cap L3 for small UPI failed debits; starts at L1 in sandbox, L2 live.	Amber tier. V 4.78 · F 3.29 · S 5.00 → priority index 4.30, rank #1.	Quick win; reshape play; Wave 1, live from month 5.

Inputs

12 months of case, alert, reconciliation and contact data; 25+ interviews; process walks; the bank's AI inventory and vendor contracts.

The funnel

50 use cases identified → 21 pass the gates and are scored → 10 prioritised → 2 pilots in the engagement.

Why this order

Type and autonomy are fixed *before* scoring, so a use case cannot score well on value it could only deliver with autonomy the bank will not allow.

The '5-lens assessment' is the method used throughout this document; lenses 4 and 5 draw on standard value–feasibility and portfolio practice.

Lens 1 decomposes payments into 10 value-chain stages across five business lines

1. Onboard	2. Initiate	3. Authenticate	4. Screen	5. Process	6. Settle	7. Reconcile	8. Resolve	9. Serve	10. Comply
KYC, KYB, merchant and client set-up	Channels, checkout, mandates, agent payments	Authentication and authorisation decisions	Fraud, AML, sanctions, scam interdiction	Routing, repair, trade and cross-border processing	Clearing, settlement, incidents	Switch, CBS, NPCI and network reconciliation	Disputes, chargebacks, complaints	Servicing, engagement, collections, monetisation	Regulatory, reporting, engineering enablers

Five business lines

UPI & retail payments	UPI (remitter, PSP and beneficiary bank), IMPS, NEFT, BBPS, NACH and AutoPay mandates, debit cards
Cards issuing	Credit and debit card acquisition, authorisation, servicing, rewards, disputes, collections
Merchant acquiring	POS, UPI QR and sound boxes, payment gateway, onboarding, settlement, chargebacks, MDR billing
Transaction banking	Cash management, corporate and SME payments, virtual accounts, collections reconciliation
Cross-border & trade	Remittances, SWIFT and ISO 20022, sanctions screening, nostro, LCs and collections

Why stages and lines, not departments

Departments hide hand-offs. A UPI dispute crosses the switch, the core banking system, NPCI's dispute system, the contact centre and finance. Mapping by stage shows where work, waiting and errors pile up, which is where agents add most value. A sixth row, 'cross-business', holds capabilities used by every line, such as AML triage or engineering copilots.

> **Output of Lens 1:** a 10 × 6 grid where every long-list idea and every rupee of value has an address (p. 18).

Lens 1 sizes each domain's value from activity data, so the number is defensible line by line

The value formula

> **Value** = volume × unit cost (or loss, or revenue) × AI-addressable share × expected improvement × adoption

Four value levers:

- **Cost to serve**: handling effort, contact deflection, engineering capacity
- **Loss avoidance**: fraud, write-offs, TAT compensation, penalties
- **Revenue**: activation, spend, new fee lines, retention
- **Risk and compliance**: audit findings and backlogs, valued only where a cost exists

Discipline: every input carries a label (P, A or C) and an owner on the bank's side who confirms it in weeks 2–4. Benchmarks are only a cross-check.

Worked example: dispute handling at Bank X ILLUSTRATIVE

Activity	Share of time (A)	Archetype	AI-addressable (A)	Weighted (C)
Intake and classification	12%	Conversational / document AI	80%	9.6%
Evidence gathering across switch, CBS, UDIR, network portals	35%	Agentic workflow	75%	26.2%
Rule check and resolution proposal	20%	Agentic workflow + rules	60%	12.0%
Customer and merchant communication	15%	GenAI drafting	70%	10.5%
Posting adjustments and closure	10%	Agentic (bounded)	60%	6.0%
Exceptions and escalations	8%	Human	10%	0.8%
Total addressable share of handling time				**65.1%**

900 FTE × ₹8.5 lakh × 65% addressable × 55% effort reduction = **₹27.4 crore** a year of handling effort, before TAT compensation and chargeback write-offs (p. 27).

Activity shares and addressability are assumptions to be replaced by time sampling in weeks 2–4. · Bank X is a fictional composite; bank figures are assumptions.

Lenses 2 and 3 fix the type of AI and its autonomy cap before anything is scored

Lens 2 · Archetype test (first 'yes' wins; ask in order)

Is the task a prediction on structured data (score, rank, forecast)?	→ A · Predictive ML
Is the main input language, documents or messages, with a person still acting?	→ B · Copilot or C · Document AI
Does the AI talk directly to customers or merchants?	→ D · Conversational
Must it plan and execute several steps across systems?	→ E · Agentic workflow
Is the payer an external AI agent acting for a customer?	→ F · Agentic commerce

If every answer is 'no', the problem probably needs process change or rules, not AI.

Lens 3 · Autonomy cap from error cost and reversibility

Error cost ↓ · Reversibility →	Easily reversed	Reversible with effort	Irreversible
Low (₹ small, no harm)	**L4**	**L3**	**L2**
Medium (₹ material or customer friction)	**L3**	**L2**	**L2**
High (funds blocked, regulatory breach)	**L2**	**L2**	**L1**

Overrides: RBI or NPCI rules can lower the cap (for example, regulatory filings stay at L1). A use case can start below its cap and move up only after its evaluation thresholds hold for 8 weeks in production.

Lens 3 applied to six decisions

Decision	Error cost	Reversibility	Cap
Credit a UPI failed debit under ₹5,000 (NPCI status confirmed)	Low	With effort	**L3**
Close a reconciliation break by posting an adjustment	Medium	Easily reversed	**L3**
Repair beneficiary details on a NEFT or SWIFT payment	Medium	With effort	**L2**
Submit a chargeback representment	Medium	With effort	**L2**
Hold a suspected scam payment	High	With effort	**L2**
Close an AML alert · file an STR	High	Irreversible	**L1**

Lens 4 scores value, feasibility and strategic fit on anchored 1–5 scales with weights agreed upfront

Dimension	Criterion	Weight	1 =	3 =	5 =
Value	**Financial impact**	25%	< ₹5 cr a year	₹15–30 cr	> ₹50 cr
Value	**Customer and merchant experience**	10%	No visible change	Faster for one segment	Step change for millions
Value	**Risk and regulatory benefit**	10%	None	Closes a known audit gap	Materially lowers fraud, AML or regulatory exposure
Feasibility	**Data availability and quality**	10%	Scattered, unlabelled	Available, needs cleaning	Clean, labelled history
Feasibility	**Integration complexity**	10%	Core switch or CBS changes	APIs exist for most systems	Standalone or API-ready
Feasibility	**Technique maturity**	8%	Frontier, unproven	Proven elsewhere	Proven at scale in banking
Feasibility	**Change effort and time to value**	7%	> 12 months, many teams	6–9 months	< 4 months, one team
Strategic fit	**Reuse of platform components**	10%	One-off build	Some reuse	Reusable pattern for 3+ use cases
Strategic fit	**Alignment with strategy and differentiation**	10%	Nice to have	Supports a stated priority	Core to the payments strategy

Priority index = 0.45 × Value + 0.35 × Feasibility + 0.20 × Strategic fit, each the weighted average of its criteria (1–5).

Governance of the scores: weights are agreed with the steering committee before scoring; scores are set in workshops with operations, risk and technology; a ±10-point weight shift is tested for rank changes.

Risk is a gate, not a weight: every use case must clear six tests before it can be ranked

	Gate	Test	Anchor
G1	**Data residency**	Does all payment system data stay in India, including prompts, logs and embeddings?	RBI, Storage of Payment System Data (April 2018)
G2	**Personal data and consent**	Is there a lawful purpose, notice and minimisation; is personal data masked where not needed?	DPDP Act 2023 and Rules (main obligations from 13 May 2027)
G3	**Customer harm and reversibility**	Is the proposed autonomy within the cap from Lens 3?	Bank conduct rules; RBI customer-service and compensation rules
G4	**Model risk**	Can it be inventoried, validated and explained, with human oversight and a kill switch?	RBI FREE-AI (Aug 2025); draft model-risk guidance (June 2026)
G5	**Third party and outsourcing**	Do vendor terms bar training on bank data and give audit, exit and incident rights?	RBI Master Direction on Outsourcing of IT Services (2023); CERT-In directions (2022)
G6	**Network and scheme rules**	Does it follow NPCI and card-network rules for disputes, TAT, chargebacks and agent payments?	NPCI procedural guidelines; RBI TAT framework for failed transactions (2019)

● Green: passes all gates
Ranked as designed.

● Amber: passes with conditions
Ranked with added controls or a lower starting autonomy.

● Red: fails as designed
Redesigned, held at L1, or sequenced last with a named risk owner.

Regulatory anchors as published by RBI, MeitY, NPCI and CERT-In; check the latest versions during the engagement.

Lens 5 turns scores into a balanced, sequenced portfolio rather than a ranked list

1 · Place on the matrix

Value against feasibility, bubble size = ₹ value, colour = risk tier. Quadrant lines at value 3.0 and feasibility 3.25 (the middle of the scored range).

- **Quick wins**: start now
- **Strategic bets**: start once platform and controls exist
- **Fill-ins and enablers**: start early if they build reusable capability
- **Later**: revisit at each wave gate

2 · Balance the portfolio

Mix three plays (BCG's deploy, reshape, invent): about 15% of value from **deploy** (copilots), 75% from **reshape** (end-to-end journeys), 10% from **invent** (new revenue). Check the archetype and autonomy mix so the bank is not betting everything on customer-facing agents.

3 · Sequence with four rules

1. Foundations first: LLM gateway, India-hosted models, data products, evaluation harness.
2. Internal before customer-facing.
3. Supervised before autonomous: move up the ladder only on evidence.
4. Prove one domain, then replicate the pattern (disputes → chargebacks → merchant disputes).

The rules applied to Bank X's shortlist

Wave	Rule that put a use case here	Use cases	₹ cr / yr
Wave 1 · M0–6	Quick wins and enablers; internal; L1–L2; build the platform	S01, S02, S03, S06, S08, S11, S17, S18	**212**
Wave 2 · M6–15	Customer- and merchant-facing, or replicates a proven Wave 1 pattern	S04, S05, S07, S09, S10, S12, S13, S15, S19	**170**
Wave 3 · M15–24	New revenue, Red tier, or waits on external rules	S14, S16, S20, S21	**84**

Output: three waves with gates at months 3, 6, 12 and 18. Each gate checks realised value, evaluation results and incidents before the next wave is funded.

Deploy, reshape, invent: BCG (2024). Shares shown are the outcome for Bank X (p. 24).

Bank X is a top-five private bank processing about 1.4 billion UPI payments a month

Profile (all A, calibrated to public market data)

Metric	Bank X	Market context (P)
Retail customers	**6.5 crore**	
UPI remitter transactions	**1.4 billion a month**	≈5.8% of India's 24.07 bn (Sept 2026)
Credit cards in force	**1.25 crore**	≈10% of India's 12.4 crore cards
Credit card spend	**₹21,000 crore a month**	≈10% of ₹2.02 lakh crore (Aug 2026)
Debit cards	**5 crore**	
Merchant acquiring	**7 lakh POS · 45 lakh UPI QR · 30,000 online merchants**	
Transaction banking	**9,000 corporate CMS clients · 28 lakh MSME current accounts**	
Cross-border & trade	**40 lakh remittances · 1.5 lakh trade transactions a year**	
Disputes & complaints (payments)	**36 lakh a year (UPI 24 lakh, cards 12 lakh)**	
AML alerts (payments-related)	**18 lakh a year, >95% closed as false positives**	

Payments cost and loss base, ₹ crore a year (A)

Payments, cards & merchant operations (6,000 FTE)	**₹540**
Contact centre: payments & cards contacts (3,500 seats)	**₹210**
Card collections agencies	**₹120**
Net fraud losses (cards, UPI, net banking)	**₹240**
TAT compensation, chargeback write-offs, suspense write-offs	**₹105**
Payments application development & testing	**₹550**
Total addressable base	**₹1,765**

Operations FTE include outsourced staff: disputes 900, reconciliation and settlement 1,100, merchant operations 1,000, financial crime 1,400, trade and cross-border 700, cards operations 900.

> **First data the diagnostic confirms:** FTE and cost by team, 12 months of dispute and alert outcomes, and fraud loss by typology. Together they drive about 70% of the value at stake.

Market context: NPCI (Sept 2026) and RBI card data (Aug 2026), as reported by Business Standard. · Bank X is a fictional composite; bank figures are assumptions.

Value concentrates where people do the work: serving, screening and resolving hold 63% of the value at stake

₹ crore a year (C)	1. Onboard	2. Initiate	3. Authenticate	4. Screen	5. Process	6. Settle	7. Reconcile	8. Resolve	9. Serve	10. Comply	Total
UPI & retail payments		4		39		4	13	40	22		123
Cards issuing		3		6		2	6	20	82		120
Merchant acquiring	24	3				2	6	7	41		83
Transaction banking		19			6		18				42
Cross-border & trade				10	20						30
Cross-business				27						40	66
Total value	**24**	**29**	**0**	**82**	**25**	**9**	**44**	**68**	**145**	**40**	**465**
Long-list use cases	5	5	2	7	4	2	4	4	11	6	50

Reading the map: darker cells hold more value. Stage 9 (serve, engage, collect) is largest because it combines servicing cost, card revenue and collections. Screening and dispute resolution follow. Authentication (stage 3) has ideas but no shortlisted value: those are classic-ML improvements already in the bank's fraud roadmap.

Breadth versus depth: ideas are spread across every stage (bottom row), but value is not. That is why the portfolio is built around domains (p. 19), not around the longest list of ideas.

Shortlisted value allocated to stages and lines using assumed splits (e.g., disputes 60% UPI, 30% cards, 10% acquiring). · Bank X is a fictional composite; bank figures are assumptions.

Five domains hold 70% of the ₹465 crore a year at stake



Top five domains (blue): Financial crime, Disputes & complaints, Customer & merchant servicing, Cards engagement & collections, Transaction banking & cross-border.

How the total is built: 21 shortlisted use cases, each sized with the value formula (appendix p. 45). Range ₹350–580 crore (±25%) reflects uncertainty in adoption and improvement rates.

Not counted: option value of agentic commerce beyond Year 3, avoided regulatory penalties without a cost history, and staff redeployed to growth roles.

Calculated from assumed drivers; annual run-rate once each use case is fully ramped. · Bank X is a fictional composite; bank figures are assumptions.

We identified 50 use cases across the value chain (1 of 2)

ID	Use case	Stage	Lines	Archetype	Cap	Shortlist
L01	Merchant onboarding and KYB agent	1. Onboard	ACQ	E · Agentic workflow	L2	S08
L02	CMS and SME onboarding document intelligence	1. Onboard	TXB	C · Document AI	L2	–
L03	Video-KYC deepfake and document-tamper detection	1. Onboard	UPI, CRD	A · Predictive ML	L3	–
L04	Vernacular re-KYC outreach agent	1. Onboard	UPI	D · Conversational	L3	–
L05	Merchant category and business-model classification	1. Onboard	ACQ	C · Document AI	L3	S08
L06	Conversational payments in Indian languages	2. Initiate	UPI	D · Conversational	L2	–
L07	Delegated mandates and agent tokens (issuer)	2. Initiate	UPI, CRD	F · Agentic commerce	L3	S21
L08	Agent-ready checkout and agent verification (acquirer)	2. Initiate	ACQ	F · Agentic commerce	L3	S21
L09	Corporate and SME payments copilot	2. Initiate	TXB	E · Agentic workflow	L2	S14
L10	ISO 20022 enrichment and remittance structuring	2. Initiate	TXB, XBT	C · Document AI	L2	–
L11	Risk-based adaptive authentication	3. Authenticate	UPI, CRD	A · Predictive ML	L4	–
L12	Issuer authorisation optimisation (false declines)	3. Authenticate	CRD	A · Predictive ML	L4	–
L13	Scam and mule-network interdiction	4. Screen	UPI	A · Predictive ML	L3	S04
L14	AML alert triage and investigation agent	4. Screen	All lines	E · Agentic workflow	L2	S02
L15	STR narrative drafting copilot	4. Screen	All lines	B · GenAI copilot	L1	S02
L16	Fraud investigation and case copilot	4. Screen	UPI, CRD	B · GenAI copilot	L2	S03
L17	Sanctions screening false-positive reduction	4. Screen	XBT	A · Predictive ML	L2	S05
L18	Merchant transaction-laundering detection	4. Screen	ACQ	C · Document AI	L2	–
L19	Synthetic-voice detection in the contact centre	4. Screen	UPI, CRD	A · Predictive ML	L3	–
L20	Smart routing and retry optimisation	5. Process	ACQ	A · Predictive ML	L4	–
L21	Payment repair agent (NEFT, RTGS, SWIFT)	5. Process	TXB, XBT	E · Agentic workflow	L3	S07
L22	Intraday liquidity and nostro forecasting	5. Process	TXB, XBT	A · Predictive ML	L1	–
L23	Payments incident triage and root-cause agent	6. Settle	UPI, CRD, ACQ	E · Agentic workflow	L2	S19
L24	Merchant settlement exception handling	6. Settle	ACQ	E · Agentic workflow	L2	–
L25	Multi-way reconciliation exception agent	7. Reconcile	UPI, CRD, ACQ	E · Agentic workflow	L3	S06

Lines: UPI = UPI and retail payments, CRD = cards issuing, ACQ = merchant acquiring, TXB = transaction banking, XBT = cross-border and trade. Cap = autonomy cap from Lens 3. · Bank X is a fictional composite; bank figures are assumptions.

We identified 50 use cases across the value chain (2 of 2); 21 pass the gates and go forward

ID	Use case	Stage	Lines	Archetype	Cap	Shortlist
L26	Corporate receivables matching	7. Reconcile	TXB	C · Document AI	L3	S13
L27	Nostro reconciliation and investigations	7. Reconcile	XBT	E · Agentic workflow	L2	–
L28	Suspense and ageing clean-up agent	7. Reconcile	UPI, CRD, ACQ	E · Agentic workflow	L2	S06
L29	UPI dispute resolution agent (UDIR)	8. Resolve	UPI	E · Agentic workflow	L3	S01
L30	Card chargeback and representment agent	8. Resolve	CRD, ACQ	E · Agentic workflow	L2	S01
L31	Complaint classification and root-cause analytics	8. Resolve	All lines	C · Document AI	L3	–
L32	Ombudsman and regulator response drafting	8. Resolve	All lines	B · GenAI copilot	L1	–
L33	Vernacular payments servicing assistant	9. Serve	UPI, CRD	D · Conversational	L3	S10
L34	Contact-centre agent-assist copilot	9. Serve	UPI, CRD, ACQ	B · GenAI copilot	L1	S11
L35	Merchant servicing agent (settlement, MDR, devices)	9. Serve	ACQ	D · Conversational	L3	S09
L36	Card collections agent	9. Serve	CRD	D · Conversational	L2	S16
L37	Corporate service-desk copilot (payment tracing)	9. Serve	TXB, XBT	B · GenAI copilot	L2	–
L38	Card servicing actions agent (limits, EMI, controls)	9. Serve	CRD	E · Agentic workflow	L3	S10
L39	Regulatory change and obligations agent	10. Comply	All lines	B · GenAI copilot	L1	S17
L40	Regulatory returns preparation copilot	10. Comply	All lines	B · GenAI copilot	L1	–
L41	Policy and SOP knowledge assistant	10. Comply	All lines	B · GenAI copilot	L1	S11
L42	Payments engineering copilot	10. Comply	All lines	B · GenAI copilot	L2	S18
L43	Partner API and developer assistant	10. Comply	ACQ, TXB	B · GenAI copilot	L3	–
L44	Card personalised offers and next-best action	9. Serve	CRD	A · Predictive ML	L3	S15
L45	Merchant insights and credit copilot	9. Serve	ACQ	B · GenAI copilot	L1	S20
L46	Trade and cross-border document intelligence	5. Process	XBT	C · Document AI	L2	S12
L47	Acquiring deal-pricing copilot (incl. UPI MDR)	9. Serve	ACQ	B · GenAI copilot	L1	–
L48	Transaction-banking RM copilot	9. Serve	TXB	B · GenAI copilot	L1	–
L49	Rewards and loyalty optimisation	9. Serve	CRD	A · Predictive ML	L3	–
L50	Audit evidence and data-localisation compliance agent	10. Comply	All lines	E · Agentic workflow	L1	–

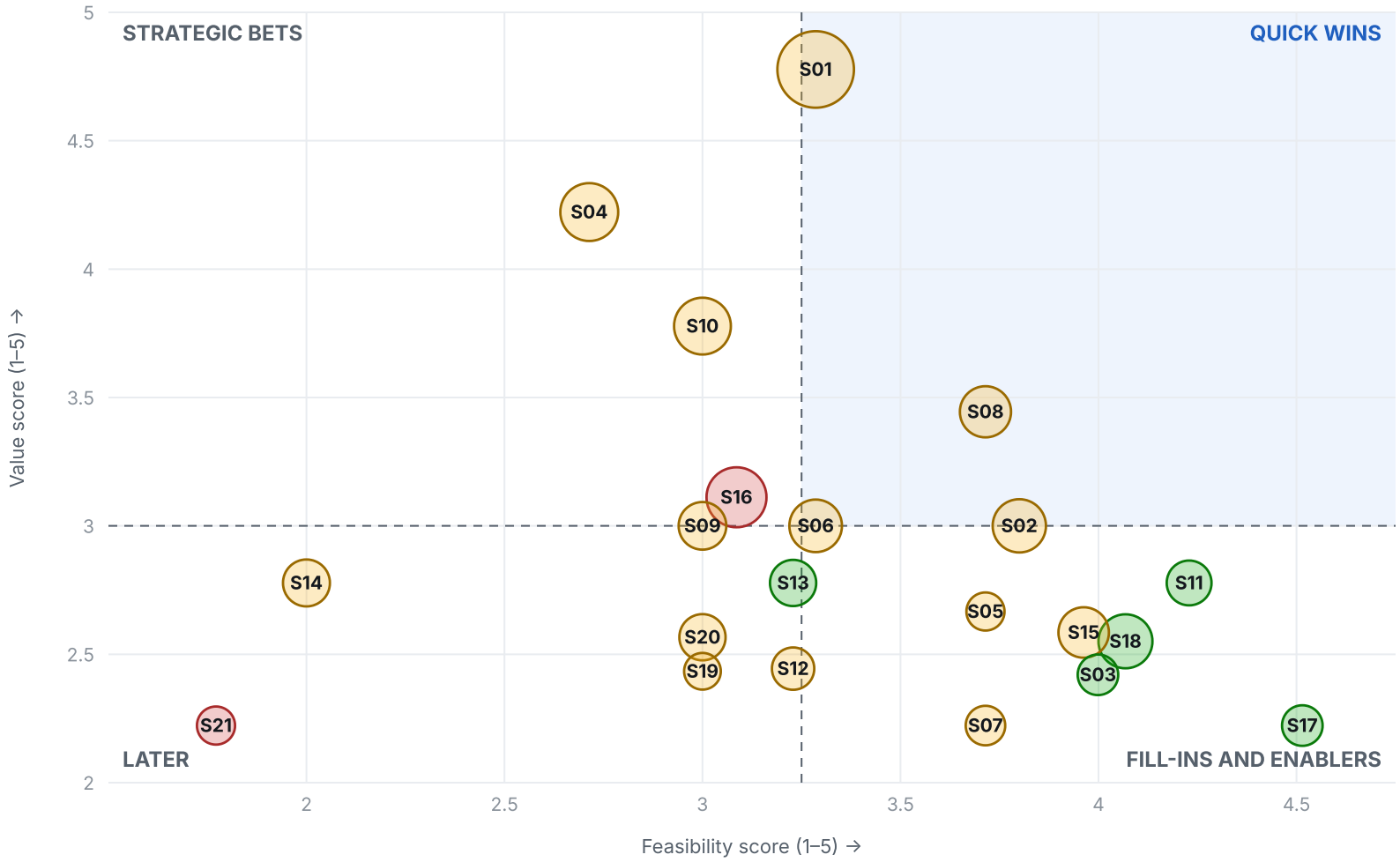
Several long-list items combine into one shortlisted use case (e.g., L29 and L30 form S01). Items without a shortlist ID are parked or folded into classic-ML roadmaps. · Bank X is a fictional composite; bank figures are assumptions.

Disputes lead the scored shortlist; servicing and scam interdiction follow despite harder delivery

ID	Use case	₹ cr / yr	Value	Feasib.	Fit	Priority index	Risk tier	Autonomy	Quadrant	Wave
S01	Dispute & chargeback agent	₹68	4.78	3.29	5.00	**4.30**	Amber	L1 → L3	Quick win	W1
S10	Vernacular servicing assistant	₹32	3.78	3.00	5.00	**3.75**	Amber	L1 → L3	Strategic bet	W2
S04	Scam & mule interdiction	₹33	4.22	2.71	4.00	**3.65**	Amber	L2 → L3	Strategic bet	W2
S08	Merchant onboarding agent	₹24	3.44	3.71	4.00	**3.65**	Amber	L1 → L2	Quick win	W1
S02	AML alert triage agent	₹27	3.00	3.80	4.00	**3.48**	Amber	L1 → L2	Quick win	W1
S11	Agent-assist copilot	₹16	2.78	4.23	3.50	**3.43**	Green	L1 → L1	Fill-in / enabler	W1
S06	Reconciliation exception agent	₹26	3.00	3.29	4.50	**3.40**	Amber	L2 → L3	Quick win	W1
S18	Engineering copilot	₹28	2.56	4.03	4.00	**3.36**	Green	L1 → L2	Fill-in / enabler	W1
S17	Regulatory change agent	₹12	2.22	4.51	3.50	**3.28**	Green	L1 → L1	Fill-in / enabler	W1
S09	Merchant servicing agent	₹19	3.00	3.00	4.00	**3.20**	Amber	L1 → L3	Strategic bet	W2
S03	Fraud investigation copilot	₹12	2.44	4.00	3.50	**3.20**	Green	L1 → L2	Fill-in / enabler	W1
S15	Card engagement	₹23	2.56	4.00	3.00	**3.15**	Amber	L2 → L3	Fill-in / enabler	W2
S05	Sanctions false-positive reduction	₹10	2.67	3.71	3.00	**3.10**	Amber	L1 → L2	Fill-in / enabler	W2
S16	Collections agent	₹37	3.11	3.09	3.00	**3.08**	Red	L1 → L2	Strategic bet	W3
S13	Receivables matching	₹18	2.78	3.23	3.50	**3.08**	Green	L2 → L3	Later	W2
S20	Merchant insights & credit	₹18	2.56	3.00	4.00	**3.00**	Amber	L0 → L1	Later	W3
S07	Payment repair agent	₹11	2.22	3.71	3.50	**3.00**	Amber	L2 → L3	Fill-in / enabler	W2
S12	Trade & cross-border documents	₹14	2.44	3.23	3.50	**2.93**	Amber	L1 → L2	Later	W2
S14	SME payments copilot	₹19	2.78	2.00	4.00	**2.75**	Amber	L1 → L2	Later	W3
S19	Incident triage agent	₹9	2.44	3.00	3.00	**2.75**	Amber	L1 → L2	Later	W2
S21	Agentic commerce	₹10	2.22	1.77	4.00	**2.42**	Red	L0 → L3	Later	W3

Scores from the Lens 4 rubric (p. 14); ties broken by ₹ value. All scores are assumptions for illustration, to be set in scoring workshops. · Bank X is a fictional composite; bank figures are assumptions.

Four quick wins start now; disputes is the biggest of them, and the customer-facing bets wait for proven controls



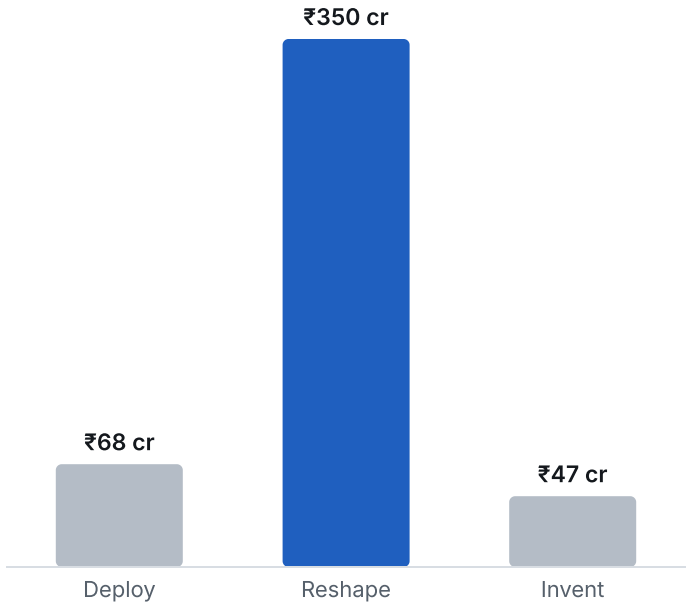
ID	Use case	₹ cr
S01	Dispute & chargeback agent	₹68
S02	AML alert triage agent	₹27
S03	Fraud investigation copilot	₹12
S04	Scam & mule interdiction	₹33
S05	Sanctions false-positive reduction	₹10
S06	Reconciliation exception agent	₹26
S07	Payment repair agent	₹11
S08	Merchant onboarding agent	₹24
S09	Merchant servicing agent	₹19
S10	Vernacular servicing assistant	₹32
S11	Agent-assist copilot	₹16
S12	Trade & cross-border documents	₹14
S13	Receivables matching	₹18
S14	SME payments copilot	₹19
S15	Card engagement	₹23
S16	Collections agent	₹37
S17	Regulatory change agent	₹12
S18	Engineering copilot	₹28
S19	Incident triage agent	₹9
S20	Merchant insights & credit	₹18
S21	Agentic commerce	₹10

● Green ● Amber ● Red risk tier · bubble area ∝ ₹ value

Bubble positions are nudged slightly where they would overlap; exact scores on p. 22. · Bank X is a fictional composite; bank figures are assumptions.

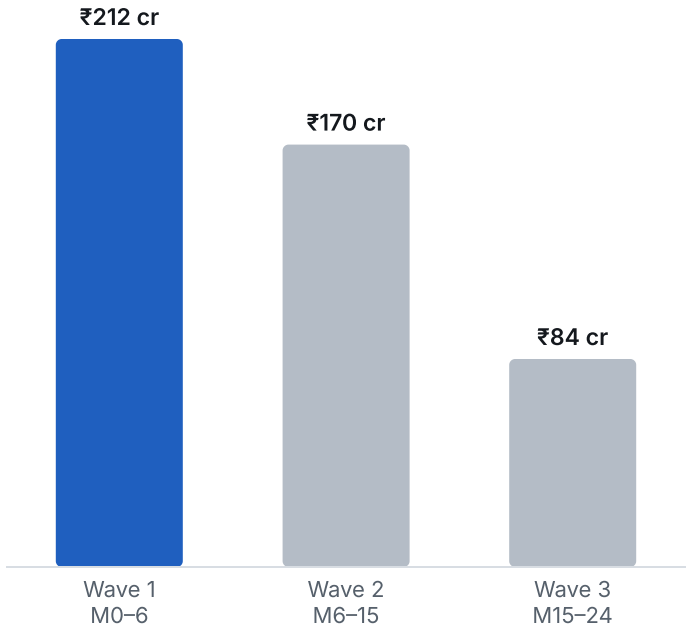
The portfolio is three-quarters 'reshape': value comes from rewiring journeys, with copilots as the on-ramp

Value by play, ₹ crore a year



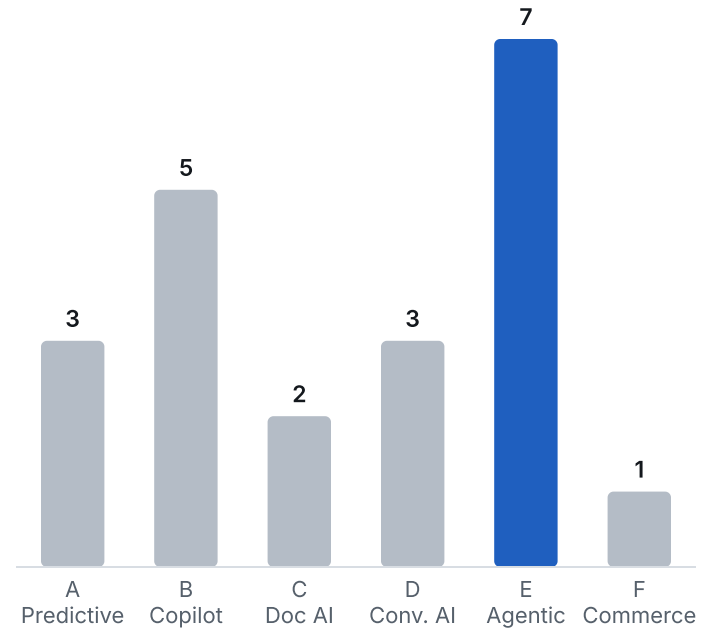
Reshape = 75%. Deploy copilots are small but build skills and platform early.

Value by wave, ₹ crore a year



Wave 1 carries 46% of value: internal agents with supervised autonomy.

Shortlisted use cases by archetype



Agentic workflows (E) are the largest group (7 of 21); classic ML and conversational AI add 6 more.

Plays follow BCG's deploy, reshape, invent classification; assignment of each use case is the practice's judgement. · Bank X is a fictional composite; bank figures are assumptions.

The top 10 use cases carry ₹284 crore a year, 61% of the value at stake

ID	Use case	What changes	₹ cr / yr	Autonomy	Wave	Business owner	Proof point in 12 weeks
S01	**Dispute & chargeback agent**	Agent gathers switch, CBS, UDIR and network evidence, proposes resolution; officer approves	**₹68**	L1 → L3	W1	Head of Payment Operations	Sandbox pilot on 500 past cases
S10	**Vernacular servicing assistant**	Customers resolve payment queries in 10+ languages, with actions such as raising a dispute	**₹32**	L1 → L3	W2	Head of Customer Service	Intent model on 3 months of chats
S04	**Scam & mule interdiction**	Graph ML spots mule networks; a conversational warning interrupts likely scams	**₹33**	L2 → L3	W2	Head of Fraud Risk	Back-test on 12 months of scam losses
S08	**Merchant onboarding agent**	Agent reads KYB documents, checks GST and Udyam data, pre-fills the case for a checker	**₹24**	L1 → L2	W1	Head of Merchant Acquiring	50-file KYB extraction test
S02	**AML alert triage agent**	Agent assembles alert context and drafts the disposition; analyst decides	**₹27**	L1 → L2	W1	Head of Financial Crime	Sandbox pilot on 1,000 closed alerts
S11	**Agent-assist copilot**	Live transcript, next-best action and auto-summary for contact-centre staff	**₹16**	L1 → L1	W1	Head of Contact Centre	Summary quality on 200 calls
S06	**Reconciliation exception agent**	Agent investigates reconciliation breaks and proposes postings within limits	**₹26**	L2 → L3	W1	Head of Reconciliation	Break-cause analysis on 1 month
S18	**Engineering copilot**	Code modernisation and test generation for NPCI and network changes	**₹28**	L1 → L2	W1	Payments CTO	Test generation on one NPCI change
S17	**Regulatory change agent**	RBI and NPCI circulars turned into an obligations register with owners	**₹12**	L1 → L1	W1	Chief Compliance Officer	Obligations register for 20 circulars
S09	**Merchant servicing agent**	Merchants get settlement, MDR billing and device help in their language	**₹19**	L1 → L3	W2	Head of Merchant Services	Top-10 merchant intents mapped

The next 11 use cases stay in the portfolio and are reviewed at each wave gate; collections (S16) and agentic commerce (S21) are Red-tier and start only with named risk owners.

Ranked by priority index; ties broken by ₹ value. · Bank X is a fictional composite; bank figures are assumptions.

Disputes: an agent that gathers evidence and proposes resolutions could cut handling cost per dispute by about a third and reduce TAT penalties

Today at Bank X (A)

- 36 lakh disputes a year: 24 lakh UPI (mostly failed or pending debits), 12 lakh card disputes and chargebacks
- 900 FTE; about ₹212 handling cost per dispute
- ₹38 crore a year of TAT compensation under RBI's failed-transaction framework
- ₹55 crore a year of chargebacks written off: evidence late or incomplete
- Staff toggle between 6–8 systems to assemble one case

> **Why agentic:** most effort is evidence gathering and rule checks across systems, which is exactly what tool-using agents do well.

Target design: dispute agent with human checkpoints

1 Intake
App, UDIR, contact centre, branch; classify reason code

→

2 Gather
Switch logs, CBS ledger, NPCI UDIR, network portals, merchant data

→

3 Check
NPCI and network rules, RBI TAT, prior outcomes (policy RAG)

→

4 Propose
Resolution, credit or representment with evidence pack

→

5 Act
Post within limits, notify customer, close and log

TOOLS VIA APIS
UPI switch, CBS, NPCI UDIR, card-network dispute systems, case management, CRM, notification service

HUMAN CHECKPOINTS
Officer approves every credit at L2; at L3, auto-resolve only UPI failed debits under ₹5,000 where NPCI status confirms; all chargeback representments reviewed

CONTROLS
Maker-checker in the ledger, ₹ and daily volume limits, full audit trail per step, kill switch, weekly sample review by quality team

One dispute, before and after (A)

Measure	Today	With the agent
Process steps	9	**4**
Hand-offs between teams	3–4	**1**
Systems opened by staff	6–8	**1 workbench**
Average resolution time	5.8 days	**under 2.5 days**
Handling cost per dispute	₹212	**about ₹136**

UDIR = NPCI's Unified Dispute and Issue Resolution. RBI TAT framework for failed transactions (Sept 2019). · Bank X is a fictional composite; bank figures are assumptions.

Disputes: ₹68 crore a year at stake, unlocked in steps from supervised to bounded autonomy

Value drivers (C from A)

Lever	Calculation	₹ cr / yr
Handling effort	900 FTE × ₹8.5 lakh × 65% addressable × 55% effort cut	**₹27.4**
TAT compensation	₹38 crore paid × 55% fewer breaches	**₹20.9**
Chargeback write-offs	₹55 crore written off × 35% recovered by complete, on-time evidence	**₹19.2**
Total		**₹67.5**

Autonomy path

Months 1–3	L1 in sandbox: agent drafts, officers compare with their own decisions
Months 4–8	L2 live for UPI disputes: agent proposes, officer approves in one click
Months 9–12	L3 for UPI failed debits under ₹5,000 with confirmed NPCI status; cards chargebacks move to L2

KPIs and evaluation thresholds

Measure	Baseline (A)	Target
Average resolution time	5.8 days	under 2.5 days
Disputes resolved without manual evidence gathering	~0%	60%
Reason-code classification accuracy (golden set)	n/a	≥ 95%
Evidence pack completeness	~70%	≥ 98%
Wrong credits (false-credit rate)	0.3%	≤ 0.1%
Agreement of policy-check judge with senior officers	n/a	≥ 90%
Ombudsman complaints on disputes	index 100	index 70

> **Go/no-go for L3:** thresholds met for 8 consecutive weeks, zero unauthorised postings, sign-off by the CRO and the model-risk team.

90-DAY PILOT PLAN

Days 1–20
Golden set of 500 closed UPI disputes, labelled by senior officers

→

Days 21–50
Build in the sandbox: tools on masked data, policy RAG, graders

→

Days 51–80
Shadow mode on live cases: agent proposes, officers decide as today

→

Days 81–90
Go/no-go against thresholds; steering decision

USE-CASE RISKS AND CONTROLS

Invented evidence	Every claim must cite a system record; grader rejects uncited claims
Wrong credit	₹ limits, maker-checker, daily reversal check
Prompt injection	Customer text treated as data; tools allow-listed per step
TAT breach if agent fails	Fallback to the manual queue at a fixed age

All baselines and targets are assumptions to be set from Bank X's case data in weeks 2–4. · Bank X is a fictional composite; bank figures are assumptions.

Financial crime: agents clear routine AML alerts with evidence while ML and conversational warnings interrupt scams

AML alert triage and investigation agent · S02 · ₹27 cr a year

Today (A): 18 lakh payments-related alerts a year, over 95% closed as false positives; 600 analysts.

Collect	→	Analyse	→	Draft	→	Decide
KYC, transactions, counterparties, adverse media		Network graph, typology match, prior STRs		Disposition and STR narrative with citations		Analyst approves; FIU-IND filing stays human

- Autonomy L1 → L2; regulatory filing never above L1
- Value: analyst productivity +45% on addressable work; backlog and external review avoided
- Evaluation: agreement with senior investigators ≥ 90%; zero missed escalations on a seeded test set

Scam and mule-network interdiction · S04 · ₹33 cr a year

Today (A): ₹240 crore net fraud loss a year; about 55% linked to scams and mule accounts that rules miss.

Detect	→	Warn	→	Pause	→	Learn
Graph ML on payee networks; DPIP and mule signals		Conversational warning in the customer's language		Short hold only within approved rules		Outcomes feed models and the registry

- Autonomy L2 for holds, L3 for warnings; friction budget agreed with the business
- Value: 25% reduction in scam- and mule-linked losses
- Guardrails: explainable reasons for each hold, fast release path, fairness checks across regions and languages

Why this pairing works: classic ML stays the real-time engine; generative AI speeds up investigations and talks to customers at the moment of risk. Shared fraud intelligence (DPIP, MuleHunter.AI) makes the signals stronger for every participating bank.

Data and systems
Transaction monitoring alerts and outcomes, KYC, case manager, adverse-media feed, payee graph, DPIP and MuleHunter.AI signals, prior STRs.

Regulatory notes
PMLA record-keeping and STR timelines still apply; the analyst remains accountable for every disposition; model-risk validation before L2.

Pilot 2 in the engagement
1,000 closed alerts replayed in the sandbox; agent dispositions compared with senior investigators; zero missed escalations required.

DPIP Smart Registry and MuleHunter.AI as reported (Sept and July 2026). FIU-IND = Financial Intelligence Unit, India. · Bank X is a fictional composite; bank figures are assumptions.

Reconciliation: an agent can clear routine breaks across switch, CBS and NPCI files within posting limits

Daily files
NPCI settlement, switch, CBS GL, card networks, PG partners

→

Match engine
Rules plus ML matching; most items auto-match

→

Break queue
Unmatched items by type and age

→

Agent investigates
Reads logs and statements; finds cause; drafts adjustment

→

Approve or auto-post
Maker-checker (L2); auto-post under ₹ limit (L3)

→

GL and customer
Posting, reversal, customer notice, root-cause log

Value · ₹37 cr a year

Reconciliation agent (S06) ₹26 cr: 700 FTE, 70% addressable, 50% effort reduction, plus 40% lower suspense write-offs. Payment repair agent (S07) ₹11 cr uses the same pattern for NEFT, RTGS and SWIFT exceptions.

KPIs

- Breaks resolved within T+1: 60% → 90%
- Suspense ageing over 30 days: −50%
- Wrong postings: zero tolerance; all reversible
- Customer reversals within RBI TAT: ≥ 99.5%

Why it ranks high

Structured data, clear rules, measurable outcomes and reuse: the same agent pattern serves disputes, payment repair and nostro investigations. It is the bank's best place to learn bounded autonomy safely.

Break taxonomy sets the autonomy, type by type (A)

Break type	Share of breaks	What the agent does	Autonomy
Timing differences (cut-off, T+1 files)	38%	Confirms the matching item in the next cycle and closes	**L3**
NPCI deemed-success versus CBS mismatch	17%	Checks UDIR and switch status; proposes credit or reversal	**L2 → L3 under ₹5,000**
Duplicate or partial postings	15%	Identifies the duplicate and drafts the reversal	**L2**
Reversal not posted within TAT	12%	Posts the reversal and the TAT compensation	**L2 → L3**
Data or format errors in files	11%	Repairs the record and re-runs the match	**L3**
Other or unexplained	7%	Summarises the evidence for an analyst	**L1**

Servicing: a vernacular assistant can resolve payment queries end to end, but only with agent tools behind it

Design

Channels
App chat, WhatsApp, IVR voice; 10+ Indian languages

→

Assistant
Authenticates, understands, plans

→

Tools
Transaction status, raise dispute, card controls, refund status, MDR bill

→

Hand-off
To a person with full context when needed

- **Customers (S10):** 'Where is my refund?', 'Money debited but not received', 'Block my card', in their own language
- **Merchants (S09):** settlement queries, device faults, and from 15 Oct 2026 questions about MDR on UPI payments above ₹2,000
- **Staff (S11):** agent assist with live transcript, next-best action and auto-summary for the calls that remain

Value and guardrails

Vernacular servicing assistant (S10)	**₹32 cr**
Merchant servicing agent (S09)	**₹19 cr**
Agent-assist copilot (S11)	**₹16 cr**

- Authenticate before any action; no financial advice; scripted refusals
- Language quality checks by native reviewers on a weekly sample
- Containment is measured with resolution, not just deflection: a contact that returns within 7 days counts as a failure
- Escalation to a person on frustration, vulnerability cues or repeated failure

> **KPIs:** resolution without a person 20% → 45%; repeat contacts −25%; CSAT parity with human agents.

Where the contacts come from, and how far the assistant can go (A)

Contact reason	Share	Tool the assistant needs	Autonomy	Contact reason	Share	Tool the assistant needs	Autonomy
Debited, not received	24%	UPI status, raise dispute	**L3**	Registration or PIN	9%	Guided reset flow	**L3**
Refund status	13%	Refund tracker	**L3**	AutoPay or mandate	7%	Mandate view, revoke	**L3**
Card block, unblock, limits	12%	Card controls	**L3**	Bill pay or FASTag	6%	BBPS and FASTag status	**L3**
Charge or fee query	11%	Fee explainer, reversal request	**L2**	Other	18%	Hand-off with context	**L1**

NPCI's MyUPI (Sept 2026) shows the direction: an AI assistant for complaints in five languages inside BHIM. · Bank X is a fictional composite; bank figures are assumptions.

Agentic commerce is an option to build now, before the liability rules settle

What is happening (P)

- Oct 2025: NPCI, OpenAI and Razorpay piloted UPI payments inside ChatGPT, using Reserve Pay limits set by the user; Axis Bank and Airtel Payments Bank were the bank partners
- Sept 2026: Reuters reported NPCI's Unified Agent Protocol for delegated agent payments; liability and limits not yet public
- Card networks run their own agent programmes

What it means for Bank X

- **As issuer:** mandates and agent tokens with limits; customer consent UX; disputes when an agent buys the wrong thing
- **As acquirer:** agent-ready checkout for merchants; verifying which agent is paying
- **Threats:** agent-driven fraud; loss of the customer relationship to the agent platform

No-regret moves (Wave 3, option earlier)

1. Join NPCI agentic pilots as issuer and acquirer
2. Build mandate and limit management with clear consent
3. Make fraud models agent-aware
4. Draft dispute rules for agent-initiated payments
5. Offer merchants an agent-readiness kit

HOW AN AGENT PAYMENT WORKS, AND WHERE BANK X SITS

1 Consent Customer sets a mandate: merchant types, ₹ limit, validity → **2 Token** Issuer binds a token to the agent and the mandate → **3 Agent pays** Agent initiates within limits; acquirer verifies the agent → **4 Authorise** Issuer checks mandate, risk and agent identity → **5 Dispute** Customer contests a purchase; agent logs as evidence

Valued conservatively: ₹10 crore of early revenue in Year 3, with option value not counted. Red tier until NPCI publishes liability rules; first pilot only with a named risk owner and capped limits.

OPTIONS FOR BANK X

Option	For	Against
Build own agent stack	Control of the experience	Cost; rules not settled
Partner and pilot now Recommended	Learning, low cost, voice in NPCI rule-making	Depends on partner roadmaps
Wait for the rules	No spend now	Peers set the customer habits first

MediaNama and Electronic Payments International (Oct 2025); Startupfeed and Inc42 (Sept 2026). · Bank X is a fictional composite; bank figures are assumptions.

Three waves over 24 months, each funded only after a gate confirms value, evaluation results and incident record



◆ Gate M3
Platform and sandbox live; governance approved; pilots pass sandbox evaluation

◆ Gate M6
Wave 1 agents live at L1–L2; value tracking running; no Sev-1 incidents

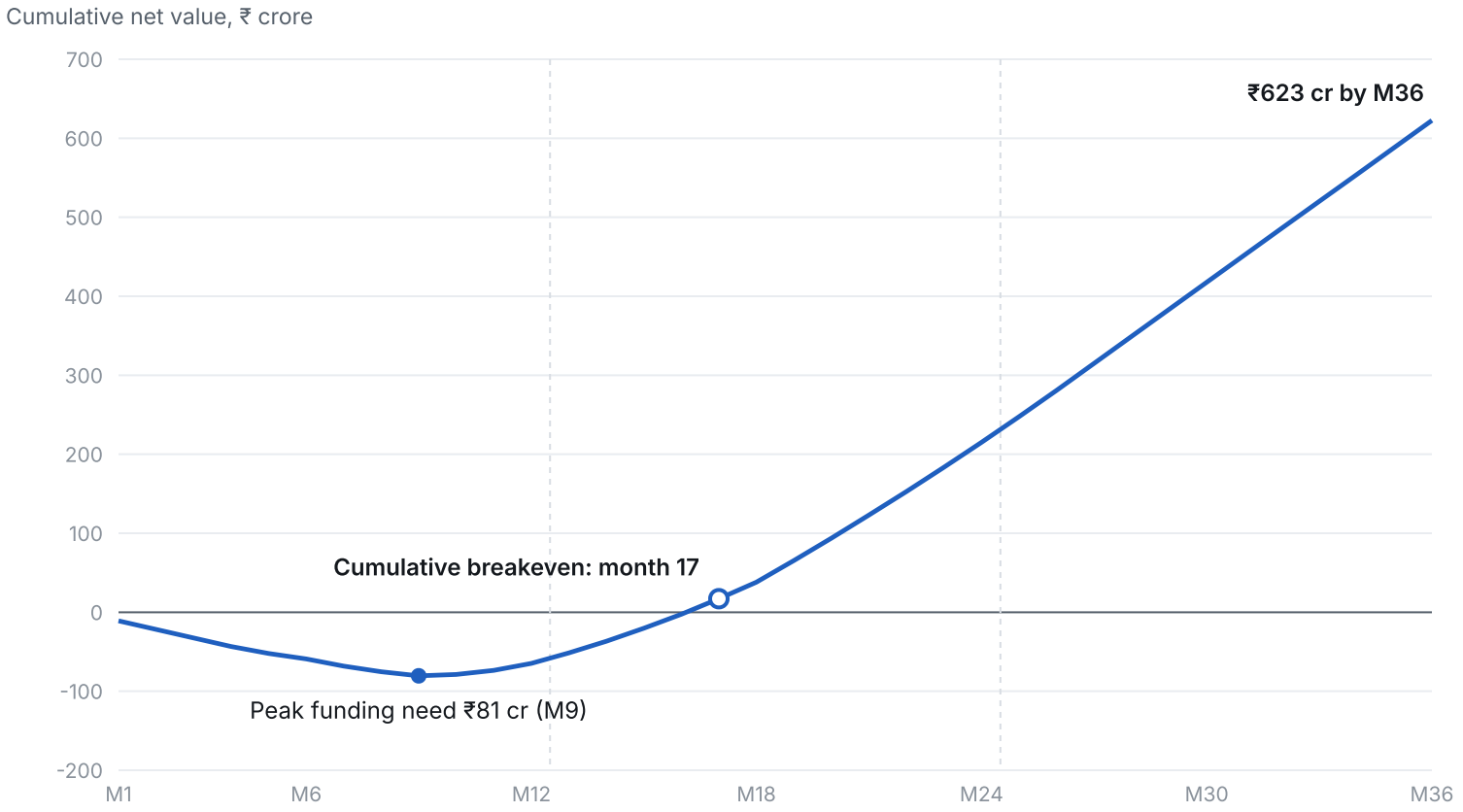
◆ Gate M12
Wave 1 at run-rate; first L3 approvals; Wave 2 customer pilots pass

◆ Gate M18
Wave 2 scaled; ₹ value at least 70% of plan; model-risk review clean

◆ Gate M24
Wave 3 decisions on collections and agentic commerce

Months from programme start. Gate checks: realised value versus plan, evaluation thresholds, incidents, regulatory feedback. · Bank X is a fictional composite; bank figures are assumptions.

The programme breaks even in month 17 and returns about ₹623 crore net over three years



Value and cost by year, ₹ crore (C)

	Value	Cost	Net
Year 1	₹75	₹140	**₹-65**
Year 2	₹365	₹85	**₹280**
Year 3	₹463	₹55	**₹408**

One-time investment (A)

AI platform and foundations	₹60	M1–M12
Wave 1 build	₹45	M1–M9
Wave 2 build	₹50	M7–M18
Wave 3 build	₹25	M13–M24
Total	**₹180**	

Run cost (A): ₹10 cr in Year 1, ₹35 cr in Year 2, ₹55 cr a year at scale (inference, hosting, platform team, evaluation).
Sensitivity: breakeven month 20 at 70% of value; month 15 at 130%.

Value ramps linearly over 9 months from each wave's go-live (months 5, 11, 19). Costs and ramps are assumptions. · Bank X is a fictional composite; bank figures are assumptions.

The target architecture keeps every model, prompt and log in India and puts one gateway between agents and models

Channels	Mobile app · WhatsApp · IVR and voice · branch and RM tools · merchant app · corporate portal · staff desktops
Agents	Dispute agent · reconciliation agent · AML triage · onboarding · servicing · copilots — orchestrated with tool permissions per agent
AI platform	LLM gateway (routing, caching, cost control) · India-hosted frontier models and open-weight models in the bank's data centre · RAG and policy knowledge · prompt, tool and agent registry
Tools and systems	UPI switch · CBS · card management · NPCI UDIR · card-network dispute systems · case management · CRM · payment hub · notification service
Data	Payments lakehouse · feature store · vector store · labelled case history · evaluation sets

CROSS-CUTTING CONTROLS

- Guardrails: input and output filters, PII masking
- Evaluation harness and red-teaming
- Observability: traces, cost, latency
- Identity, access and tool permissions
- Audit trail and kill switch
- Model inventory and validation (MRM)

Hosting: Indian cloud regions and the bank's data centres only, per RBI's payment-data storage directive. Vendor-neutral: the gateway lets Bank X switch models without rebuilding agents.

FIVE DESIGN PRINCIPLES

1 · India only
Models, prompts, logs and embeddings stay in Indian regions.

2 · One gateway
Every model call passes one point for policy, cost and logging.

3 · Least-privilege tools
Each agent gets only the APIs and ₹ limits its task needs.

4 · Evaluate before release
No prompt, model or tool change ships without passing its tests.

5 · Log everything
Every step, tool call and decision is traceable for audit.

Reference architecture; product choices are made in the engagement against the evaluation criteria, not assumed.

A hub-and-spoke operating model: a central CoE sets standards and runs the platform; domain pods own the outcomes

AI & Payments CoE (hub)

- Head of AI for payments (reports to Head of Payments, dotted line to CDO)
- Platform engineering: gateway, orchestration, data products
- Evaluation and model-risk liaison
- Responsible AI and policy

Domain pods (spokes)

- One pod per domain: product owner from operations, 2–3 AI engineers, a process designer, a data analyst, a risk partner
- Own outcomes and KPIs for their domain
- Release through the CoE's evaluation gate

Risk, compliance and audit

- CRO owns model-risk framework and autonomy approvals
- Compliance maps RBI and NPCI obligations
- Internal audit reviews agent audit trails quarterly

Talent plan (A)

Role	Month 6	Month 18
CoE platform and evaluation	18	30
Domain pods (6–8 people each)	4 pods · 28	8 pods · 56
Responsible AI, risk and compliance	5	9
Total	**51**	**95**

People who do the work today

Operations staff move from doing every step to supervising agents, handling exceptions and improving rules. Capacity released is redeployed first to backlog, quality and growth roles; reductions happen through attrition, with a published plan agreed with HR. This is where the 70% of the effort in the 10-20-70 rule goes.

HOW AN AGENT GOES LIVE

1 Propose
Pod writes the use-case card: value, cap, tools

→

2 Gate check
Six risk gates; tier set

→

3 Build
Tools, prompts, golden set, graders

→

4 Validate
Model-risk review; red-team

→

5 Approve
CRO signs autonomy level

→

6 Run
Live monitoring; quarterly review

Team sizes are assumptions for a bank of Bank X's scale. · Bank X is a fictional composite; bank figures are assumptions.

Governance follows RBI's FREE-AI pillars, so the programme is ready for supervisory review from day one

FREE-AI pillars → what Bank X puts in place

Infrastructure	India-hosted models and gateway; shared evaluation data; sandbox with masked production data
Policy	Board-approved AI policy covering generative and agentic AI; autonomy approval process
Capacity	AI literacy for the board and senior management; training for operations supervisors of agents
Governance	AI inventory including vendor models; named owners; risk tiering; steering and gate reviews
Protection	Customer disclosure when talking to AI; human hand-off; grievance redress; DPDP consent and notices
Assurance	Independent validation; kill switches; incident reporting (including CERT-In within 6 hours); quarterly audit of agent trails

Other rules that shape the design

Storage of Payment System Data (RBI, April 2018)	All payment data in India: prompts, logs, embeddings, model outputs
Draft model-risk guidance (RBI, June 2026)	Board-approved framework, inventory, validation, explainability, human oversight, kill switches; third-party models included
DPDP Act 2023 and Rules (Nov 2025)	Notice, consent and purpose; main obligations from 13 May 2027
Outsourcing of IT Services (RBI, 2023)	Vendor due diligence, audit and exit rights for AI vendors
NPCI and card-network rules	Dispute TATs, chargeback rules, agent-payment rules as they are issued

FREE-AI (Framework for Responsible and Ethical Enablement of AI) sets 7 sutras, 6 pillars and 26 recommendations. The June 2026 guidance was a consultation draft as of our sources; the design assumes it is finalised broadly as drafted.

KPMG India summary of the FREE-AI report (Aug 2025); S&R Associates and CorpLawUpdates on the draft model-risk guidance; Legal500 on DPDP Rules.

Value is tracked at three levels, from visible change in weeks to financial results in months

Level	Measures	Owner and cadence
Visible change	Board-approved AI policy and autonomy process; two agents live; disputes and reconciliation teams working with agents daily	Steering committee
Interim (operational)	Dispute resolution time; breaks cleared within T+1; AML alerts closed per analyst-day; containment with resolution; evaluation pass rates; incidents	Domain product owners, weekly
Financial	Handling cost per case; TAT compensation; chargeback write-offs; fraud loss; contact cost; new revenue lines	Finance, monthly against the value plan

Value plan: each use case has a baseline from Bank X's own data, a target and a finance owner. Benefits count only when they show up in a budget line, a loss line or revenue.

Evaluation is part of the KPI set: every agent has a golden test set, automated graders calibrated against senior staff, and online quality metrics. A model or prompt change ships only if evaluations hold.

Example: tracking the disputes value (A)

Value line	Baseline source	Measured by	Counts as value when	₹ cr / yr at stake
Handling effort	Ops FTE and time sampling	Head of Payment Operations	Headcount or outsourcing budget reduced, or backlog absorbed with no new hiring	27.4
TAT compensation	GL line for TAT payouts	Finance	Monthly payouts fall against a volume-adjusted baseline	20.9
Chargeback write-offs	Chargeback loss GL	Cards finance	Win rate on representments rises; write-offs fall	19.2

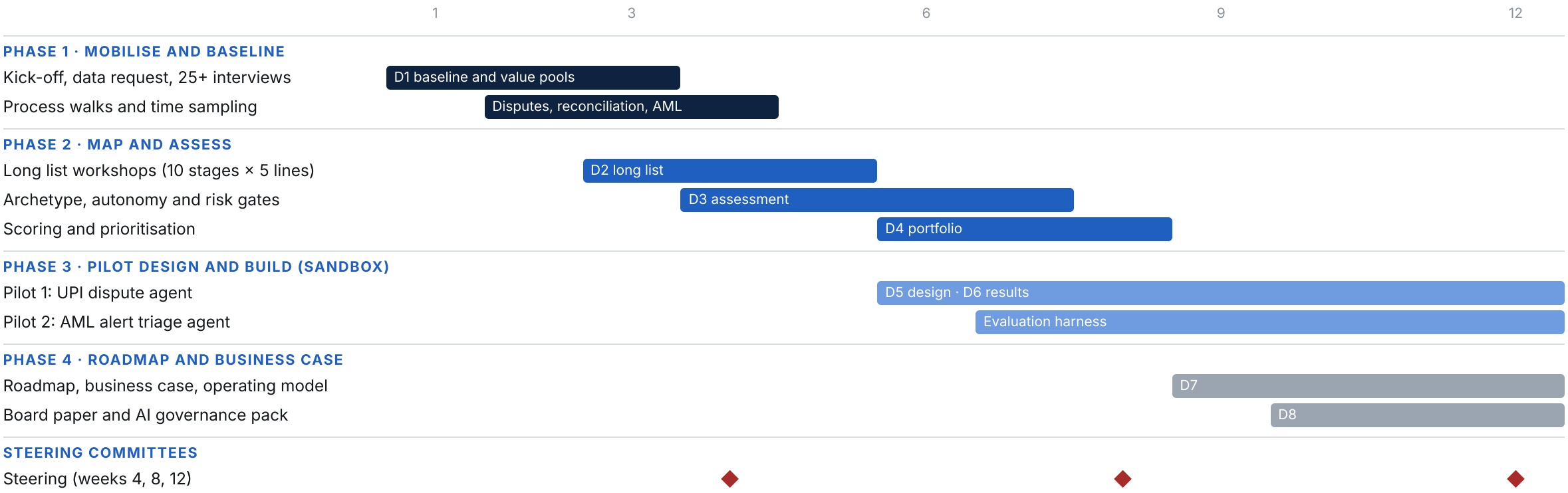
Seven programme risks, each with a mitigation built into the plan rather than added later

Risk	Rating	Mitigation in the plan
Value not realised: capacity freed but not released	Medium	Value plan signed by finance; redeployment plan agreed with HR before go-live
Agent error reaches customers	Medium	Autonomy caps, maker-checker, evaluation thresholds, kill switch, staged rollout
Regulatory change mid-programme	Medium	FREE-AI-aligned governance; modular design; quarterly regulatory review
Vendor lock-in or model price changes	Medium	LLM gateway, at least two model options per use case, open-weight fallback
Data access and quality delays	High	Data request on day 2; masked sandbox by week 3; data products built in Wave 1
Change resistance in operations	High	Operations leaders own pods; supervisors trained first; visible quick wins
Inference cost overrun	Low	Routing to smaller models, caching, cost budgets per agent

Watch list for the first 90 days: sandbox and masked data ready by week 3; data request complete by day 5; final RBI model-risk guidance and any NPCI agent-payment circulars; vendor terms on training with bank data; staff consultation and redeployment plan agreed with HR.

Ratings are the practice's initial view, to be confirmed with Bank X's risk function.

A 12-week diagnostic ends with a validated portfolio, two sandbox pilots and a go or no-go decision



Deliverables: D1 baseline and value pools (wk 3) · D2 long list (wk 5) · D3 assessment and gate results (wk 7) · D4 prioritised portfolio (wk 8) · D5 pilot designs and evaluation harness (wk 9) · D6 pilot results (wk 12) · D7 roadmap, business case, operating model (wk 12) · D8 board paper and governance pack (wk 12).

Go/no-go at week 12: proceed to Wave 1 if the diagnostic confirms at least ₹150 crore a year of validated Wave 1 value and both pilots meet their evaluation thresholds with zero unauthorised actions in the sandbox.

The first week is planned day by day, and the bank's commitments are named upfront

Week 1

Day 1	Kick-off with the sponsor and steering members; confirm objectives, measures and governance; issue the data request
Day 2	Interviews: heads of disputes, reconciliation, cards operations and merchant operations; review the AI inventory
Day 3	Interviews: CRO, CISO, Chief Compliance Officer, head of financial crime; data-localisation and vendor review
Day 4	Process walks on the disputes desk and in reconciliation; time sampling; contact-reason analysis
Day 5	Synthesis: first value pools and hypotheses; sandbox plan; sponsor check-in

> **Enabling commitment:** the steering committee agrees in week 1 to act on the week-12 go/no-go, whichever way it goes.

Data request, issued on day 1

Data (12 months unless stated)	Bank X owner	Due
Dispute cases with reason codes, outcomes and TAT	Head of Payment Operations	Day 5
Reconciliation breaks by type, age and resolution	Head of Reconciliation	Day 5
AML alerts with dispositions and STRs filed	Head of Financial Crime	Day 5
Contact reasons and volumes by channel	Head of Customer Service	Day 5
Operations FTE and cost by team, incl. outsourced	Finance business partner	Day 5
Fraud loss by typology and channel	Head of Fraud Risk	Day 5
Merchant onboarding TAT and rework	Head of Merchant Acquiring	Day 10
AI inventory, cloud regions, vendor contracts	CIO and CDO	Day 10
Sandbox with masked production data	CIO and CISO	Week 3

A senior, payments-specialist team, with the bank's people working alongside in every workstream

Our team (Option B, 12 weeks)

Role	Days	Name
Partner, Payments Practice Lead	24	[TO CONFIRM: named individual]
Associate Partner, AI & Data	30	[TO CONFIRM: named individual]
Engagement Manager	60	[TO CONFIRM: named individual]
Senior Consultant	120	[TO CONFIRM: named individual]
Consultant	60	[TO CONFIRM: named individual]
AI / ML Architect	36	[TO CONFIRM: named individual]
Data Scientist	60	[TO CONFIRM: named individual]
Payments Operations SME	30	[TO CONFIRM: named individual]
Risk & Regulatory Expert	18	[TO CONFIRM: named individual]

Credentials: [TO CONFIRM: one relevant payments-AI case example, with client permission].

Bank X team alongside us

Role	Time
Executive sponsor (Head of Payments)	2 hours a week; chairs steering
Programme lead	50% for 12 weeks
Six domain SMEs (disputes, reconciliation, financial crime, servicing, merchant, cards)	4 hours a week each
Data engineer	50% for 12 weeks
Risk and model-risk partner	3 hours a week
Legal and compliance reviewer	2 hours a week

Engagement governance

Steering committee (weeks 4, 8, 12)

Head of Payments (chair), CRO, CIO, Chief Compliance Officer, Head of Operations, CDO; our Partner and Associate Partner.

Working groups (weekly)

Value and domains · Risk and regulation · Technology and data · People and change. Each pairs a bank lead with a team member.

Quality and independence

A second partner not on the team reviews D4 and D7. We do not resell technology; any vendor shortlist is built against criteria agreed with Bank X.

Days per role from the fee build on p. 42.

Fees: ₹6.88 crore for the recommended 12-week option, about 68 times smaller than the annual value at stake

Fee build, Option B (A: day rates are placeholders)

Role	Days	₹ lakh / day	₹ crore
Partner, Payments Practice Lead	24	4.0	0.96
Associate Partner, AI & Data	30	3.0	0.90
Engagement Manager	60	1.8	1.08
Senior Consultant	120	1.2	1.44
Consultant	60	0.9	0.54
AI / ML Architect	36	1.6	0.58
Data Scientist	60	1.0	0.60
Payments Operations SME	30	1.4	0.42
Risk & Regulatory Expert	18	2.0	0.36
Total professional fees	**438**		**6.88**

Options

Option	Scope	₹ crore
A · Focused	8 weeks; three domains (disputes, reconciliation, financial crime); one pilot design	3.58
B · Standard Recommended	12 weeks; full value chain; two sandbox pilots; roadmap and business case	6.88
C · Extended	B plus 12 weeks of support to take both pilots live	11.40

- Expenses at actuals, capped at 8% of fees (Option B: ₹0.55 crore)
- GST at 18% extra
- Payment: 30% at mobilisation, 40% at week 8, 30% at week 12
- Offer valid for 60 days
- Value-to-fee: ₹465 crore a year at stake ÷ ₹6.88 crore ≈ 68×

Day rates are illustrative placeholders to be replaced by the firm's rate card; fees are built bottom-up as days × rate. · Bank X is a fictional composite; bank figures are assumptions.

Compliance with the RFP: nine requirements met in full and two partly, with each deviation explained

Ref	Requirement (assumed RFP)	Where answered	Compliance
R1	Understanding of objectives and scope	p. 4–5	**Full**
R2	Assessment framework and methodology	p. 10–16	**Full**
R3	Use cases across the payments value chain	p. 18–21	**Full**
R4	Prioritisation and roadmap	p. 22–25, 32	**Full**
R5	Value case and investment	p. 19, 33, 45	**Full: indicative until validated in weeks 2–4**
R6	Regulatory alignment (RBI, NPCI, DPDP)	p. 15, 36	**Full**
R7	Target architecture, vendor-neutral	p. 34	**Full: criteria provided; no vendor selection**
R8	Operating model and governance	p. 35–36	**Full**
R9	Pilots within the engagement	p. 39–40	**Partial: sandbox pilots; production go-live in Option C**
R10	Team and credentials	p. 41	**Partial: names and permitted credentials to confirm**
R11	Commercial proposal	p. 42	**Full**

R9, partial: pilots run in a sandbox on masked data within the 12 weeks. Production go-live needs Bank X's change, security and model-risk approvals, so it sits in Option C rather than being promised inside the diagnostic.

R10, partial: named team members and a client-permitted case example are confirmed at submission. No credentials or references are claimed in this draft.

We red-teamed this response as the bank's panel would; the weakest spot is team credentials, which must be confirmed

Simulated evaluation panel: not buyer feedback

Criterion (assumed)	Weight	Score	Reason
Understanding of requirements	15%	**8/10**	Playback is specific; open questions show judgement
Methodology and framework	30%	**8/10**	Five lenses and gates are clear; value method is defensible
Regulatory and risk approach	15%	**8/10**	Strong RBI mapping; final MRM rules still pending
Team and experience	20%	**6/10**	Names and credentials not yet confirmed
Commercials	20%	**7/10**	Bottom-up and phased; Option B above some rivals' price
Weighted score		**7.4/10**	

Three strongest reasons to choose someone else, and our answer

An auditor-led rival claims regulatory comfort
Our answer: risk gates and FREE-AI mapping built into the method (pp. 15, 36); independent validation stays with the bank's model-risk team.

A technology vendor offers a free diagnostic bundled with a build
Our answer: vendor-neutral architecture and value validated on Bank X's data; pilots in 12 weeks without a platform commitment.

The bank's own AI CoE could do it
Our answer: co-delivery with the CoE, cross-bank payments patterns, and a handover plan from week 9.

Changes made after the review: added the day-by-day first week, autonomy caps with a red line, and a smaller Option A. Still open: named team and a permitted case example.

Scores are a simulated review by the authoring team, labelled as such; they are not evidence of how a real panel would score.

APPENDIX ILLUSTRATIVE

Appendix: how each use case's value is calculated (1 of 2)

ID	Use case	Driver and calculation (A inputs, C results)	₹ cr / yr
S01	**Dispute & chargeback agent**	Handling effort: 900 FTE × ₹8.5 lakh × 65% addressable × 55% effort cut = ₹27.4 TAT compensation: ₹38 crore paid × 55% fewer breaches = ₹20.9 Chargeback write-offs: ₹55 crore written off × 35% recovered by complete, on-time evidence = ₹19.2	**₹67.5**
S02	**AML alert triage agent**	Analyst productivity: 600 analysts × ₹11 lakh × 70% addressable × 45% faster = ₹20.8 Backlog and external review avoided: Assumed ₹6 crore a year = ₹6.0	**₹26.8**
S03	**Fraud investigation copilot**	Investigator productivity: 400 investigators × ₹11 lakh × 70% × 40% = ₹12.3	**₹12.3**
S04	**Scam & mule interdiction**	Fraud loss avoided: ₹240 crore net loss × 55% scam/mule-linked × 25% reduction = ₹33.0	**₹33.0**
S05	**Sanctions false-positive reduction**	Screening effort: 200 FTE × ₹11 lakh × 80% × 40% = ₹7.0 Faster cross-border release: Assumed fee and float benefit = ₹3.0	**₹10.0**
S06	**Reconciliation exception agent**	Break investigation effort: 700 FTE × ₹8.5 lakh × 70% × 50% = ₹20.8 Suspense write-offs: ₹12 crore × 40% = ₹4.8	**₹25.6**
S07	**Payment repair agent**	Repair effort: 400 FTE × ₹8.5 lakh × 60% × 45% = ₹9.2 Penalties and fee leakage: Assumed = ₹2.0	**₹11.2**
S08	**Merchant onboarding agent**	Onboarding effort: 500 FTE × ₹8.5 lakh × 70% × 50% = ₹14.9 Faster activation revenue: 9 lakh new merchants × 4 pts more activated × ₹2,500 revenue = ₹9.0	**₹23.9**
S09	**Merchant servicing agent**	Servicing effort: 500 FTE × ₹8.5 lakh × 70% × 45% = ₹13.4 Lower merchant attrition: Assumed = ₹6.0	**₹19.4**
S10	**Vernacular servicing assistant**	Contact deflection: ₹210 crore × 65% addressable × 30% deflected = ₹40.9 Run cost (models, voice): Assumed = ₹-9.0	**₹31.9**
S11	**Agent-assist copilot**	Handle-time reduction: Remaining ₹169 crore × 80% adoption × 12% AHT = ₹16.2	**₹16.2**

FTE costs: ₹8.5 lakh (operations), ₹11 lakh (financial-crime specialists), ₹10 lakh (trade), ₹18 lakh (compliance). All inputs assumed for illustration. · Bank X is a fictional composite; bank figures are assumptions.

APPENDIX ILLUSTRATIVE

Appendix: how each use case's value is calculated (2 of 2)

ID	Use case	Driver and calculation (A inputs, C results)	₹ cr / yr
S12	**Trade & cross-border documents**	Document-check effort: 700 FTE × ₹10 lakh × 50% × 40% = ₹14.0	**₹14.0**
S13	**Receivables matching**	Mandates won and retained: Assumed fee and float = ₹14.0 Matching effort: Assumed = ₹4.0	**₹18.0**
S14	**SME payments copilot**	New SME revenue: 28 lakh MSMEs × 1.5% adopt × ₹4,500 a year = ₹18.9	**₹18.9**
S15	**Card engagement**	Spend uplift: ₹2.5 lakh crore spend × 0.6% uplift × 1% net yield = ₹15.0 Dormant-card activation: Assumed = ₹8.0	**₹23.0**
S16	**Collections agent**	Agency cost: ₹120 crore × 60% × 30% = ₹21.6 Credit cost avoided: Assumed better early-bucket cure = ₹15.0	**₹36.6**
S17	**Regulatory change agent**	Compliance productivity: 150 FTE × ₹18 lakh × 30% = ₹8.1 Penalties avoided: Assumed = ₹4.0	**₹12.1**
S18	**Engineering copilot**	Engineering capacity realised: ₹550 crore × 50% addressable × 20% × 50% realised = ₹27.5	**₹27.5**
S19	**Incident triage agent**	Downtime and decline cost: ₹30 crore baseline × 30% = ₹9.0	**₹9.0**
S20	**Merchant insights & credit**	Merchant credit margin: 45 lakh merchants × 0.4% × ₹2.5 lakh × 4% margin = ₹18.0	**₹18.0**
S21	**Agentic commerce**	Early agent-payment revenue (Year 3): Assumed; option value not counted = ₹10.0	**₹10.0**
	Total, all 21 use cases		**₹464.9**

Run-rate once fully ramped. Negative drivers are run costs netted against value. · Bank X is a fictional composite; bank figures are assumptions.

Appendix: glossary

Agentic AI	AI that plans and executes multi-step tasks using tools, within permissions and limits
APP scam	Authorised push payment fraud: the customer is tricked into sending money
CBS	Core banking system
CMS	Cash management services for corporate clients
DPDP	Digital Personal Data Protection Act 2023 and Rules 2025
DPIP	Digital Payments Intelligence Platform: shared fraud-intelligence infrastructure led by RBI
FIU-IND	Financial Intelligence Unit, India; receives suspicious transaction reports (STRs)
FREE-AI	RBI's Framework for Responsible and Ethical Enablement of AI (Aug 2025)
KYB	Know your business: verification of merchants and corporate clients
LLM gateway	A single control point between applications and models for routing, cost, logging and policy
MDR	Merchant discount rate: the fee a merchant pays on a payment
MRM	Model risk management
P2M	Person-to-merchant payment
RAG	Retrieval-augmented generation: answering from the bank's own documents
Reserve Pay	UPI feature that blocks funds for a merchant or agent within user-set limits
TAT	Turnaround time; RBI sets TATs and compensation for failed transactions
UDIR	NPCI's Unified Dispute and Issue Resolution for UPI
UAP	Unified Agent Protocol: NPCI framework for agent-initiated UPI payments, as reported (Sept 2026)

Appendix: sources and labels

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Labels used

P	Public source, cited above
A	Assumed for this illustration; to be replaced with Bank X data
C	Calculated from P and A inputs (model available on request)

How sources were used. Facts were gathered on 13 October 2026 from search summaries of the pages listed; the pages could not be opened directly from the authoring environment. Verify against the primary RBI, NPCI and MeitY releases before relying on any figure.

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